Discover Mental Health

RESEARCH Open Access

Influence of digital social networking platforms on mental health and social media addiction

Jonathan Martínez-Líbano[1], Talía Gómez Yepes[2], Andrea Sarmiento Passalacqua[3], Miguel Ángel Albalá Genol[2*], Christian A. Verdugo[1], Catalina Coronado-Reyno[1] and Edgardo Etchezahar[2]

*Correspondence:
Miguel Ángel Albalá Genol
miguel.albala@uam.es
[1]Andrés Bello National University, Santiago, Chile
[2]Department of Evolutionary and Educational Pshycholoy (Education Section), Autonomous University of Madrid, Madrid, Spain
[3]Universidad de las Américas, Santiago, Chile

Abstract

The use of social media has been associated with various effects on mental health and addiction to these platforms, especially among university students. The aim of the study was to analize the impact of Instagram, Facebook, WhatsApp, and TikTok on social media addiction and mental health, considering the mediating role of the latter. The study was conducted with a sample of 684 university students. Through Model 4 of the PROCESS macro for SPSS, direct and indirect relationships were evaluated between platform use (independent variables), mental health problems (mediating variable), and social media addiction (dependent variable). Regression analyses and bootstrapping with 10,000 samples were applied to estimate confidence intervals of indirect effects. The findings revealed Instagram and TikTok showed significant direct effects on social media addiction as well as indirect influences through mental health problems. These visually intensive platforms are associated with increases in psychological problems, which in turn contribute to higher levels of addiction. In contrast, Facebook and WhatsApp did not show significant direct or indirect effects on mental health or social media addiction. The findings highlight that platforms such as Instagram and TikTok pose significant risks to mental health and social media addiction among university students, underscoring the need for targeted interventions to mitigate these effects. On the other hand, Facebook and WhatsApp appear to have limited impact. This study fills gaps in the literature by differentiating psychological effects by platform and emphasizes the importance of considering social media design in digital well-being strategies.

Keywords Social network, Mental health, Social media, Emotional distress, Addiction

1 Introduction

Social media are interactive technologies that allow users to create and share content through networking sites or virtual communities [8]. They have become an important mechanism for human interaction in various contexts and spaces of socialization and information exchange [7]. In recent years, a range of regulatory reforms have been promoted at the international level to govern the exchange of ideas and content in digital environments, in response to growing concern about the effects of social media on psychological well-being European Commission, [15]. However, the frequent use of these

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

platforms in the daily lives of university students has raised increasing concern about their effects on mental health. Research reveals a significant correlation between social media addiction and increased levels of anxiety and depression in this population [10]. Social media addiction is considered as an inability to control the use of contact media and networks employed for socializing, which may lead to clinical conditions and deterioration of individual and interactional behaviour, as well as problems in personal functioning [7].

Problematic interaction with social media is also defined as excessive engagement with these digital platforms, to the point where other activities such as education, work, interpersonal relationships, mental health, and/or well-being are impaired [6, 18]. Maladaptive social media use is directly related to a lack of self-regulation of the time spent on these platforms or to their use as a means of regulating negative moods [29]. Within this framework, problematic engagement with digital platforms can be conceptualized as the result of an interaction between individual psychological vulnerabilities and platform-specific design features aimed at sustaining user engagement. In this context, freedom of expression in digital environments is influenced not only by regulatory or contextual factors [28], but also by behavioral mechanisms associated with diminished self-regulation and loss of control, which may shape the frequency, modality, and emotional tone of users' online expression. Empirical evidence indicates that design elements such as intermittent reinforcement, quantified social feedback (e.g., likes and views), and algorithm-driven content personalization are associated with increased compulsive use and reduced autonomy in decision-making [16]. It is important to note that these mechanisms operate differently across platforms. Visually oriented and rapid-consumption platforms (e.g., Instagram, TikTok and X) tend to intensify social comparison and external validation processes, thereby increasing psychological distress and reinforcing addictive patterns of use. These dynamics may indirectly constrain expressive autonomy by promoting emotionally reactive and validation-dependent forms of participation [25]. By contrast, platforms primarily oriented toward instrumental communication (e.g., Whatsapp and some applications within Facebook) show more limited associations with addictive use and emotional dysregulation.

Data indicate that a considerable percentage of university students present moderate to severe addiction to these platforms, with usage patterns including more than nine daily checks and average addiction scores of 17.5 out of 30 [8]. Research by Dung et al. [10] shows that the majority of students use social media less than 5 h a day (68.1%). Another study by Coyne and Wooddruff [8] concluded that nearly half (44.47%) check social media 9 or more times a day or with each notification. The differential impact of each platform on psychological well-being presents particular characteristics that deserve specific analysis.

In the case of TikTok, users with addictive usage patterns exhibit deteriorated mental health outcomes, showing higher levels of stress and loneliness compared to non-users [5]. A positive correlation has also been documented between addiction to this platform and academic procrastination in young adults, suggesting that excessive use interferes with educational responsibilities [22]. The study by Dung et al. [10] indicates that frequent TikTok use correlates significantly with increased anxiety (rho = 0.002), depression (rho = 0.023), and stress (rho = 0.001) in university students. Due to the addictive

nature of the platform globally, 33.47% of university students report high levels of procrastination, which negatively affects their academic performance.

Instagram use shows particularly problematic associations with emotional well-being, with 70% of young adults reporting increases in anxiety, depression, and feelings of inadequacy [21]. The platform's emphasis on idealized images contributes significantly to body dissatisfaction and decreased self-esteem, effects that intensify proportionally to the time of exposure [2, 31]. Similarly, 74% of respondents feel pressured to look a certain way or achieve a specific lifestyle based on Instagram content [21].

Frequent use of WhatsApp shows positive correlations between its compulsive use and various mental health problems, including depression and social anxiety [19]. The addictive pattern of this application can significantly disrupt personal life, affecting both academic performance and social engagement [9]. A particularly revealing finding is that young adults are willing to make significant sacrifices to maintain their WhatsApp use, highlighting the addictive nature of the platform [8]. According to Dhiman and Varma [9], when close friends stop sending messages, 90% of students feel depressed, and nearly the same percentage (88%) feel hopeless when they lose their mobile phone.

Finally, Facebook use shows significant associations with negative mental health outcomes, manifesting in higher levels of depression and anxiety, as well as significantly lower grades in the last year [24]. Application of the Bergen Facebook Addiction Scale (BFAS) has shown that higher addiction scores correlate with lower self-esteem and greater feelings of loneliness [27]. In the study by Mohamed et al. [24], 48.8% of students reported that their studies were affected by excessive Internet use, and among them, 96.2% of Facebook addicts reported that their study hours were negatively impacted.

This overview highlights the urgent need to understand the specific mechanisms through which different social media platforms impact the mental health of university students, with the aim of developing effective intervention strategies and promoting healthier use of these technologies [7, 21].

The main aim of this study was to examine the relationship between the use of different social media platforms (Facebook, Instagram, WhatsApp, TikTok, and X/Twitter), emotional distress, and social media addiction among university students, with a specific focus on the mediating role of emotional distress in this relationship. Unlike previous research that has approached social media use in an aggregated manner, this study contributes to the existing literature by analyzing the psychological impact of each platform separately, allowing for the identification of platform-specific risk patterns associated with their distinctive features and usage dynamics. In addition, this study aims to generate relevant empirical evidence in the Latin American context, particularly among Chilean university students, thereby expanding current knowledge of this phenomenon in underrepresented sociocultural settings. The findings are expected to have both theoretical and applied implications for the development of mental health prevention and intervention strategies and digital well-being initiatives tailored to specific social media platforms.

H: There is a significant relationship between the type of social media platform used and the level of social media addiction, mediated by emotional distress in university students.

2 Method

2.1 Study design

A quantitative, cross-sectional, correlational, and explanatory design was employed in this study to analyze the direct and indirect associations (Fig. 1) between the use of different social media platforms, emotional distress, and social media addiction in a sample of Chilean university students.

This design allows for the identification of patterns of association between variables at a given point in time, which is particularly suitable for assessing statistical mediation models in complex social and psychological contexts. The choice of a cross-sectional correlational approach is justified by the exploratory nature of the phenomenon under study, as well as by the interest in examining differences between specific digital platforms (Facebook, Instagram, WhatsApp, TikTok, and Twitter) in their relationship with mental health indicators.

This approach also allows for the comparison of direct and indirect effects through mediation models, aligning with the explanatory objectives of the study.

Data collection was conducted remotely through an online questionnaire hosted on Google Forms, which was distributed via institutional and personal social media channels. Participants voluntarily provided informed consent before beginning the questionnaire, ensuring compliance with ethical principles for research involving human subjects. This study was developed in accordance with the methodological recommendations of the STROBE Statement (Strengthening the Reporting of Observational Studies in Epidemiology), which guides transparency and quality in the presentation of observational health studies.

2.2 Participants

The study sample consisted of 684 Chilean university students, with a mean age of 26.82 years (SD = 8.81) and a median age of 24 years. The sample was predominantly female, with 454 women (66.4%), 226 men (33.0%), and 4 participants identifying as another gender (0.6%). In terms of nationality, 99.3% of participants were Chilean and 0.7% were foreign.

Regarding academic progress, 25.3% of students were in their first year, 23.8% in their second year, 17.8% in their third year, 17.7% in their fourth year, 7.0% in their fifth year,

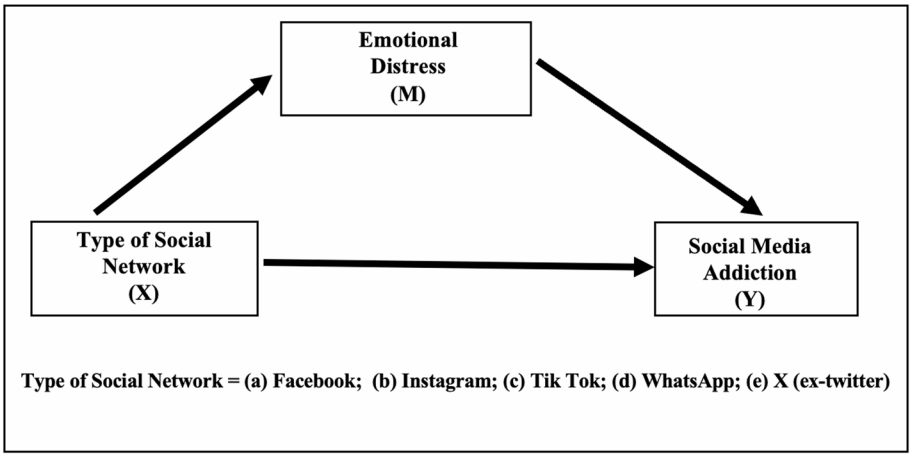


Fig. 1 Mediation model representing the main study hypothesis to be tested. Source: own elaboration

and 8.3% were in the graduation process. These data reflect a balanced distribution of students across the different stages of university education. Sampling was non-probabilistic, by convenience, through the distribution of a digital form on social media. Participants were included if they met the following criteria: being of legal age, enrolled in a higher education institution in Chile, and voluntarily agreeing to participate in the study by digitally signing the informed consent form. Individuals who did not complete the questionnaire in full or did not accept the ethical terms of participation were excluded.

2.3 Instruments

Three instruments were used to collect data, all in digital format, to evaluate social media use, emotional distress, and social media addiction. Their psychometric and operational characteristics are described below:

Questionnaire on variables associated with social media use: An ad hoc questionnaire was developed specifically for this study to assess digital behaviour variables related to social media use. The instrument was structured into four main sections. The first section collected information on social media presence, including the number of social networking platforms used by each participant (e.g., Instagram, TikTok, X/Twitter, Facebook). The second section assessed platform preference and intensity of use, asking participants to indicate the platforms they used most frequently and to identify their primary social media platform. The third section evaluated time-related usage patterns, including average daily time spent on social media (with response categories ranging from "less than 30 minutes" to "more than 4 hours per day"). The fourth section focused on usage frequency and checking behaviour, with items assessing how often participants checked social media throughout the day (e.g., several times per day, hourly, continuously). Most items were formatted using multiple-choice or ordinal response scales, depending on the nature of the variable assessed. The variables derived from this questionnaire were used to characterize participants' social media usage profiles and were subsequently included as predictor variables in the statistical analysis models.

Social Media Addiction Scale (SMAS), Chilean version: The adapted and validated version for the Chilean population of the Social Media Addiction Scale (SMAS), originally developed by Tutgun-Ünal and Deniz [30] and adapted by Miranda et al. [23], was used. The scale consists of 20 items distributed across four dimensions: social media addiction, excessive use, negative impact on personal and social life, and persistent desire to reduce use. Responses are recorded on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The Chilean version reports high internal consistency, with a Cronbach's alpha of 0.91 for the total scale.

Depression, Anxiety, and Stress Scale (DASS-21): To evaluate emotional distress, the Chilean version of the Depression Anxiety Stress Scales-21 (DASS-21), developed by Lovibond and Lovibond [20] and validated locally for the university population, was used. This instrument consists of 21 items divided into three subscales: depression, anxiety, and stress, with seven items each. Responses are recorded on a four-point Likert scale ranging from 0 (never) to 3 (almost always). The version used has shown adequate reliability levels in the Chilean population, with alpha coefficients above 0.80 in each subscale.

2.4 Procedure

Data collection was carried out between April and May 2025, remotely, using a digital questionnaire implemented through Google Forms. The link to the form was distributed via social media and institutional communication channels of different Chilean universities. This strategy enabled the inclusion of a diverse sample of higher education students from various regions of the country. The estimated completion time was approximately 12 to 15 min. To avoid duplication, responses were limited to one per email address. No financial or academic incentives were offered for participation. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki and complied with Chilean regulations for research involving human subjects.

2.5 Data analysis

Statistical analysis was performed using IBM SPSS Statistics software, version 28.0, complemented with the PROCESS macro version 4.0 developed by Andrew Hayes. A significance level of $p < .05$ was set for all inferential tests. First, descriptive analyses were conducted to characterize the sample, including measures of central tendency (mean, median), dispersion (standard deviation), and relative and absolute frequencies for categorical variables. Subsequently, correlational analyses were applied to explore bivariate relationships among the main study variables. To evaluate the direct and indirect relationships between social media platforms, emotional distress, and social media addiction, multiple mediation analysis was conducted using Model 4 of the PROCESS macro. In this model, the use of each social media platform (Facebook, Instagram, WhatsApp, TikTok, and Twitter) was included as an independent variable, emotional distress as a mediating variable, and social media addiction as the dependent variable.

The significance of indirect effects was assessed using the bootstrapping procedure with 10,000 resamples, obtaining 95% confidence intervals. Mediation was considered significant when the confidence interval of the indirect effect did not include zero. This approach allows for robust estimates of mediating effects, even in samples that do not fully meet normality assumptions.

Additionally, unstandardized coefficients (B), standard errors (SE), *p*-values, and confidence intervals were reported for each path of the model, distinguishing between direct, indirect, and total effects.

3 Results

Descriptive and comparative analyses were conducted to examine differences in emotional distress and social media addiction according to gender, age, and educational level (see Table 1). Regarding gender, statistically significant differences were observed in emotional distress ($F_{(2,\ 681)} = 7.379$, $p < .001$), with women reporting higher levels ($M = 18.96$, $SD = 16.05$) compared to men ($M = 14.51$, $SD = 14.12$). No significant differences were found in social media addiction ($F_{(2,\ 681)} = 0.143$, $p = .867$).

With respect to age, significant differences were identified in both emotional distress, $F_{(4,\ 679)} = 8.325$, $p < .001$, and social media addiction ($F_{(4,\ 679)} = 7.826$, $p < .001$). Adolescents presented the highest values of emotional distress ($M = 20.42$, $SD = 14.79$), while early older adulthood reported the lowest levels ($M = 7.45$, $SD = 13.14$). In terms of social media addiction, adolescents also obtained the highest scores ($M = 32.53$, $SD = 8.55$), in contrast to early older adulthood, which showed the lowest levels ($M = 23.95$, $SD = 6.93$).

Table 1 Descriptive statistics and analysis of differences in emotional distress and social media addiction according to sociodemographic variables

		n (%)	Emotional distress (M ± SD)	*F*	Social media addiction (M ± SD)	F
Gender	Woman	454 (66.4%)	18.96 ± 16.05	*F* = 7.379***	31.00 ± 8.69	*F* = 0.143
	Man	226 (33.0%)	14.51 ± 14.12		30.65 ± 8.42	
	Other	4 (0.6%)	29.00 ± 13.39		31.75 ± 9.91	
Age (years)	Adolescence	151 (22.1%)	20.42 ± 14.79	*F* = 8.325***	32.53 ± 8.55	*F* = 7.826***
	Emerging adulthood	264 (38.6%)	19.73 ± 15.47		31.85 ± 7.46	
	Young adulthood	160 (23.4%)	15.41 ± 16.27		29.67 ± 8.95	
	Middle adulthood	87 (12.7%)	12.45 ± 13.78		29.10 ± 10.18	
	Early older adulthood	22 (3.2%)	7.45 ± 13.14		23.95 ± 6.93	
Educational level	First year	173 (25.3%)	16.43 ± 16.20	*F* = 1.656	30.46 ± 9.16	*F* = 1.216
	Second year	163 (23.8%)	17.85 ± 14.10		31.07 ± 8.53	
	Third year	122 (17.8%)	15.31 ± 15.14		29.80 ± 7.91	
	Fourth year	121 (17.7%)	18.83 ± 16.79		31.62 ± 8.11	
	Fifth year	48 (7.0%)	18.25 ± 13.85		30.52 ± 10.04	
	Graduation process	57 (8.3%)	21.50 ± 16.72		32.74 ± 8.03	

M = mean; *SD* = standard deviation; *t* = t-test for dichotomous variables; *F* = one-way ANOVA for variables with more than two categories

***$p < .001$

Table 2 Differences in emotional distress and social media addiction according to the use of different digital platforms

		n (%)	Emotional distress		Social media addiction	
			M ± SD	t/F	M ± SD	t/F
Facebook	No	454 (66.4%)	18.17 ± 15.56		30.85 ± 8.43	
	Yes	230 (33.6%)	16.33 ± 15.54	1.459	30.96 ± 8.95	−0.149
Instagram	No	128 (18.7%)	14.36 ± 15.15		28.73 ± 8.88	
	Yes	556 (81.3%)	18.28 ± 15.59	*−2.582***	31.38 ± 8.46	*−3.175***
WhatsApp	No	250 (36.5%)	18.74 ± 15.45		30.86 ± 8.15	
	Yes	434 (63.5%)	16.86 ± 15.61	1.519	30.90 ± 8.85	−0.063
TikTok	No	475 (69.4%)	15.82 ± 15.26		29.95 ± 8.60	
	Yes	209 (30.6%)	21.48 ± 15.58	*−4.437****	33.02 ± 8.23	*−4.358****
X (Ex-Twitter)	No	624 (91.2%)	17.13 ± 15.32		30.82 ± 8.63	
	Yes	60 (8.8%)	21.93 ± 17.46	*−2.291**	31.63 ± 8.30	−0.703

M = mean; SD = standard deviation; *t* = *t*-test for dichotomous variables; F = one-way ANOVA for variables with more than two categories

***$p < .001$; **$p < .01$

Regarding educational level, no significant differences were found in either emotional distress, $F_{(5, 678)} = 1.656$, $p = .143$, or social media addiction, $F_{(5, 678)} = 1.216$, $p = .300$.

Comparisons of emotional distress and social media addiction were conducted based on the use of different digital platforms (see Table 2). No significant differences were found in emotional distress or social media addiction between Facebook users and non-users, $t_{(682)} = 1.459$, $p = .145$, and $t_{(682)} = -0.149$, $p = .881$, respectively.

In the case of Instagram, significant differences were observed in both emotional distress ($t_{(682)} = -2.582$, $p = .010$) and social media addiction ($t_{(682)} = -3.175$, $p = .002$). Instagram users reported higher levels of emotional distress ($M = 18.28$, $SD = 15.59$) and social media addiction ($M = 31.38$, $SD = 8.46$) compared to non-users.

For WhatsApp, no statistically significant differences were identified in either emotional distress ($t_{(682)} = 1.519$, $p = .129$), or social media addiction ($t_{(682)} = -0.063$, $p = .950$).

Regarding TikTok, significant differences were found in both emotional distress ($t_{(682)} = -4.437$, $p < .001$) and social media addiction ($t_{(682)} = -4.358$, $p < .001$). Users of this platform showed higher levels of emotional distress ($M = 21.48$, $SD = 15.58$) and social media addiction ($M = 33.02$, $SD = 8.23$) compared to non-users.

Finally, in the case of X (formerly Twitter), significant differences were also found in emotional distress ($t_{(682)} = -2.291$, $p = .022$) with users reporting higher levels ($M = 21.93$, SD = 17.46) than non-users ($M = 17.13$, $SD = 15.32$). However, no differences were observed in social media addiction ($t_{(682)} = -0.703$, $p = .482$).

3.1 Mediation analysis

Mediation analyses were conducted using PROCESS model 4 (Hayes, 2017) to evaluate the role of emotional distress in the relationship between the use of different digital platforms and social media addiction.

For Facebook (Fig. 2), no significant effects were observed in the model. Facebook use did not predict emotional distress ($\beta = -1.84$, $p = .145$) or social media addiction directly ($\beta = 0.51$, $p = .426$). The total effect was also not significant ($\beta = 0.10$, $p = .881$), and the indirect effect mediated by emotional distress was not significant ($\beta = -0.41$, 95% *CI* [−0.95, 0.14]).

For Instagram, both direct and indirect effects were significant. Instagram use was positively associated with greater emotional distress ($\beta = 3.92$, $p = .010$) and with social media addiction directly ($\beta = 1.81$, $p = .020$). The total effect was significant ($\beta = 2.66$, $p = .002$). Moreover, a significant positive indirect effect mediated by emotional distress was found ($\beta = 0.85$, 95% *CI* [0.21, 1.53]), indicating partial mediation (Fig. 3).

For WhatsApp, no significant associations were found (Fig. 4). Use of the application did not predict emotional distress ($\beta = -1.88$, $p = .129$), social media addiction directly ($\beta = 0.46$, $p = .467$), or total addiction ($\beta = 0.04$, $p = .950$). The indirect effect was also not significant ($\beta = -0.41$, 95% *CI* [-0.95, 0.12]).

For TikTok, results were consistent with a partial mediation model (Fig. 5). Use of this platform predicted higher levels of emotional distress ($\beta = 5.66$, $p < .001$) and social media addiction both directly ($\beta = 1.88$, $p = .005$) and totally ($\beta = 3.07$, $p < .001$). In addition, a significant indirect effect mediated by emotional distress was identified ($\beta = 1.19$, 95% *CI* [0.63, 1.85]).

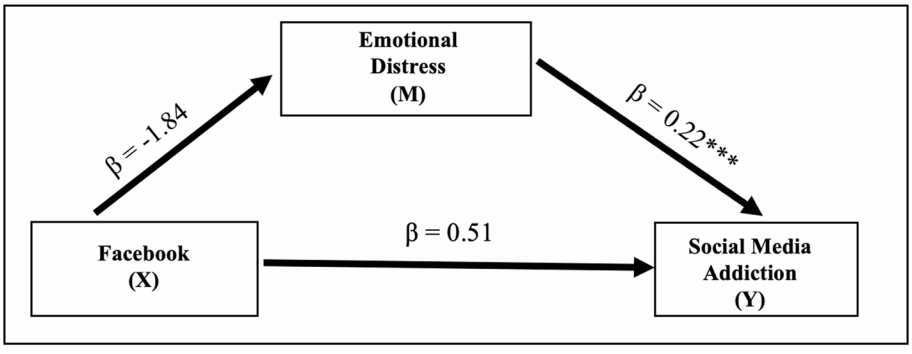


Fig. 2 Mediation model of the effect of Facebook use on social media addiction through emotional distress

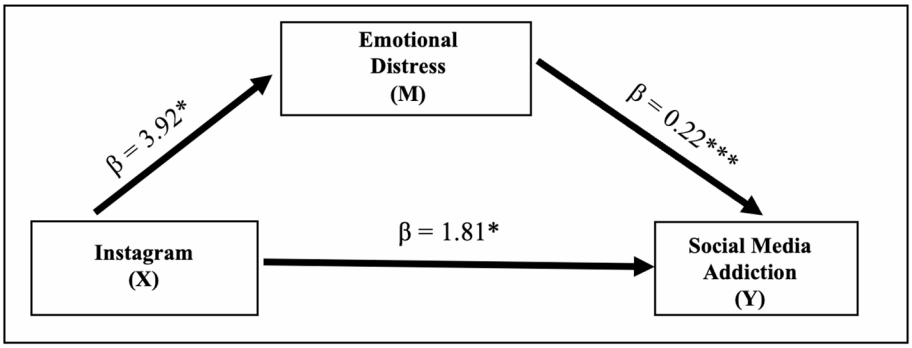


Fig. 3 Mediation model of the effect of Instagram use on social media addiction through emotional distress

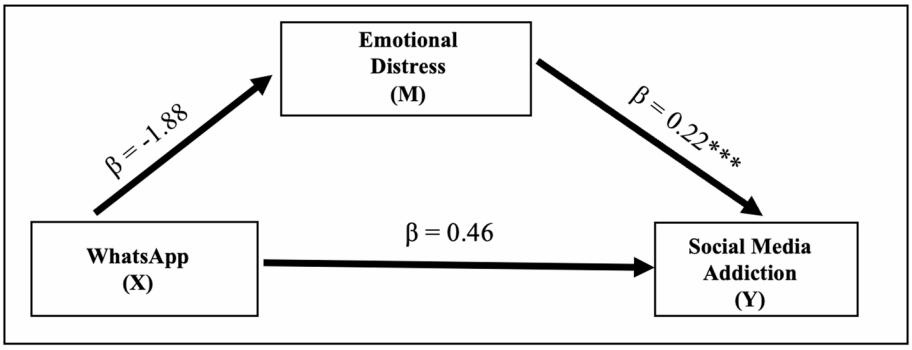


Fig. 4 Mediation model of the effect of WhatsApp use on social media addiction through emotional distress

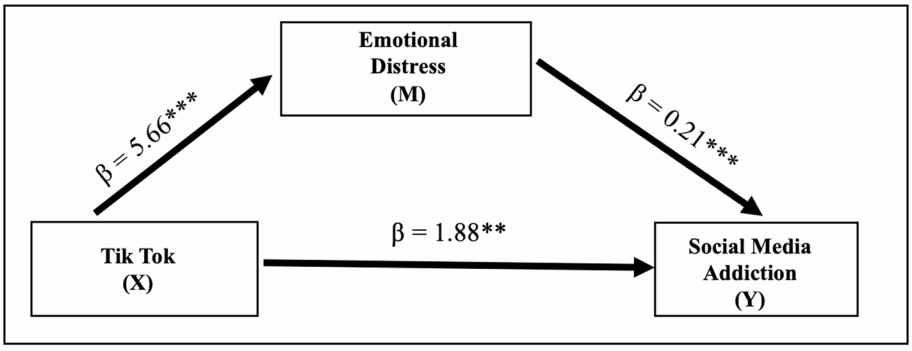


Fig. 5 Mediation model of the effect of TikTok use on social media addiction through emotional distress

Finally, in the case of Twitter (X), platform use was associated with higher levels of emotional distress ($\beta = 4.81$, $p = .022$), although no significant direct effect was found on social media addiction ($\beta = -0.24$, $p = .822$), nor on the total effect ($\beta = 0.82$, $p = .480$). However, a significant positive indirect effect through emotional distress was found ($\beta = 1.06$, 95% *CI* [0.05, 2.14]), suggesting a mediating role of this variable (Fig. 6).

Overall, the results show that emotional distress functions as a relevant mediator in the relationship between the use of certain platforms (especially Instagram, TikTok, and Twitter) and social media addiction, while for others (Facebook and WhatsApp), no significant effects are observed (Table 3).

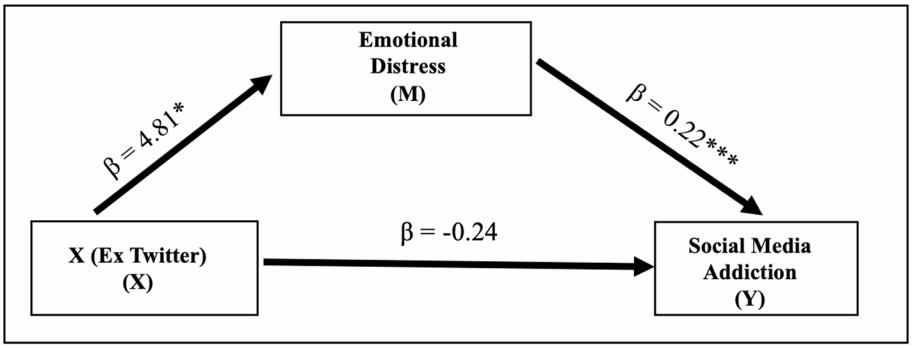


Fig. 6 Results of the mediation model for the effect of Twitter on social media addiction, mediated by emotional distress ($N = 684$)

Table 3 Results of mediation models of social media use on social media addiction, mediated by emotional distress

Plataform	a (Use → Distress)	b (Distress → Adiction)	C´(Direct)	C(Total)	a × b (Indirect)
Facebook	−1.84 ($p = .145$)	0.22*** ($p < .001$)	0.51 ($p = .426$)	0.10 ($p = .881$)	−0.41 (IC95% [−0.95, 0.14])
Instagram	3.92* ($p = .010$)	0.22*** ($p < .001$)	1.81* ($p = .020$)	2.66** ($p = .002$)	0.85 (IC95% [0.21, 1.53])
WhatsApp	−1.88 ($p = .129$)	0.22*** ($p < .001$)	0.46 ($p = .467$)	0.04 ($p = .950$)	−0.41 (IC95% [−0.95, 0.12])
TikTok	5.66*** ($p < .001$)	0.21*** ($p < .001$)	1.88** ($p = .005$)	3.07*** ($p < .001$)	1.19 (IC95% [0.63, 1.85])
Twitter (X)	4.81* ($p = .022$)	0.22*** ($p < .001$)	−0.24 ($p = .822$)	0.82 ($p = .480$)	1.06 (IC95% [0.05, 2.14])

a = effect of platform use on emotional distress; b = effect of emotional distress on social media addiction; c′ = direct effect of platform use on addiction controlling for distress; c = total effect; a × b = indirect effect calculated using bootstrapping (10,000 samples). β = unstandardized coefficient; 95% CI = 95% confidence interval

*** $p < .001$; ** $p < .01$; * $p < .05$

4 Discussion and conclusions

The findings of this study show that the repercussions of social media use on addiction vary significantly depending on the platform, supporting the need to analyze their effects separately. In particular, Instagram and TikTok exhibited significant direct and indirect effects on social media addiction, mediated by emotional distress. This result aligns with previous research demonstrating the relationship between intensive use of visually oriented platforms and an increased risk of anxiety, depression, and digital addiction [17].

These findings suggest that mechanisms of social comparison and exposure to aspirational content are key determinants of psychological vulnerability among university students, who may be particularly susceptible to the psychological conditions associated with social media addiction [4].

For TikTok, the results are consistent with recent studies highlighting the addictive nature of its rapid, personalized consumption format [22]. The observed direct and indirect associations in this platform may point to a potentially bidirectional relationship between emotional distress and addictive behaviors, although causal inferences cannot be drawn from the present analyses. Similarly, Instagram showed a pattern of partial mediation, suggesting that beyond generating emotional distress, the platform exerts a direct influence on addiction. These findings are especially relevant given that both applications are among the most widely used digital environments by university students and concentrate practices of exposure and idealization of daily life [10].

In contrast, the results for Facebook and WhatsApp did not show significant direct or indirect effects on social media addiction. One possible explanation lies in the generational shift in usage patterns: Facebook has lost centrality as a space for social interaction among young people, being used mainly for passive information consumption [27]. In the case of WhatsApp, its predominantly instrumental and communicative nature may explain why it is not associated with higher emotional distress or addictive behaviors, unlike platforms where self-exposure and social comparison are more prominent [9, 19].

The case of Twitter (now X) presented a particular pattern: the effect was indirect and mediated by emotional distress, without direct effects on addiction. This can be interpreted in light of studies showing how exposure to polarized discourse and negative news on this network contributes to adverse emotional states, which in turn increase vulnerability to problematic use [26]. However, the absence of a direct effect may indicate that, compared to Instagram or TikTok, this platform functions more as a source of emotional stress rather than as an inherently addictive environment.

From an applied perspective, the results emphasize the importance of designing intervention strategies tailored to each platform. For Instagram and TikTok, prevention should focus on critical digital literacy, self-image management, and the development of emotional competencies that help students cope with negative social comparisons [2, 13, 21]. In contrast, for platforms like Twitter, interventions should focus on managing exposure to negative content and strengthening emotional resilience [26].

These findings contribute evidence on social media use among university students and its repercussions, considering that since 2013 the *Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association, [3] has included the category "Other disorders due to addictive behaviors," encompassing gambling addiction. Therefore, given the growing evidence on problematic social media use, it would be advisable to consider social media addiction within this category to support the development of university policies and digital well-being programs that address the specific impact of each platform.

From a psychological perspective, the findings may also be interpreted through the interaction of cognitive, emotional, and behavioral components involved in social media use [1, 14]. At the cognitive level, problematic engagement with digital platforms may be associated with attentional biases, intrusive thoughts related to online content, and maladaptive beliefs regarding social validation and availability. At the emotional level, increased use may function as a coping strategy aimed at regulating negative affect, such as stress, loneliness, or anxiety, reinforcing patterns of emotional dependence on digital interaction. At the behavioral level, repetitive checking behaviors, loss of control over usage time, and habitual platform engagement may reflect compulsive tendencies that sustain problematic use patterns. Although these dimensions were not directly assessed in the present study, the observed associations between platform use, emotional distress, and indicators of social media addiction are consistent with integrative psychological models that conceptualize addictive behaviors as the result of reciprocal influences between cognition, emotion, and behavior. Future research could explicitly incorporate these components by examining cognitive distortions related to online self-presentation, emotional regulation strategies linked to platform use, and behavioral reinforcement mechanisms specific to each social media environment.

The findings of this study should be interpreted within the limits of its observational design, which focuses on the associations between the use of major social media platforms (Instagram, Facebook, WhatsApp, and TikTok), indicators of social media addiction, and mental health–related variables. Rather than suggesting direct restrictions or institutional constraints, the results point to behavioral and emotional patterns that may influence how university students interact with digital environments. In this regard, problematic engagement with social media may be associated with reduced self-regulation, attentional difficulties, and emotional vulnerability, which can indirectly shape online participation [11].

From this perspective, the potential impact on digital rights within the Chilean university context should be understood as psychosocial rather than structural or legal. Difficulties related to emotional distress or compulsive platform use may limit students' capacity to engage critically with online information, express subjective viewpoints, or sustain balanced digital communication practices. These findings underscore the relevance of considering mental health and digital well-being as key conditions supporting meaningful and autonomous participation in online academic and social spaces [12].

In this regard, the study presents some limitations that must be considered when interpreting the results. First, the cross-sectional design prevents the establishment of causal relationships between platform use, emotional distress, and social media addiction; thus, the directionality of the observed effects cannot be determined. Second, the exclusive reliance on self-report measures may have introduced social desirability bias or recall errors, limiting the accuracy of the measurements. Furthermore, the sample was obtained through convenience sampling of Chilean university students, which restricts the generalizability of the findings to other populations or sociocultural contexts. Finally, although the mediation model employed allowed exploration of relevant explanatory pathways, it did not consider other potentially influential variables, such as self-esteem, emotional regulation strategies, or perceived social support, which could enrich the understanding of the phenomenon.

Based on these limitations, future research could adopt longitudinal designs to examine how the relationships between platform use, emotional distress, and addiction evolve over time. It would also be pertinent to incorporate mixed methodologies that include both objective measures of usage time (e.g., digital records) and qualitative interviews to deepen the understanding of users' subjective experiences. Expanding the sample to include other age groups, cultural contexts, and professional backgrounds would strengthen the external validity of the findings.

It is also relevant to explore the moderating role of psychological variables such as resilience, self-control, and digital identity, which may shape the relationship between social media use and its effects on mental health. Finally, the results open the possibility of designing intervention programs and digital literacy campaigns differentiated by platform, enabling the implementation of more specific and effective preventive strategies to mitigate the impact of social media addiction in vulnerable populations such as university students.

Abbreviations

M	Mean
SD	Standard deviation
t	*t*-test for dichotomous variables
F	One-way ANOVA for variables with more than two categories

CI Confidence interval
β Beta coefficient

Author contributions
All authors have read and approved the manuscript.Authors´ contributions: J.M.L.: in the design and planning of the research within its theoretical framework (Conceptualization); in recruitment, data collection, and data cleaning and analysis; collaborating on the literature review (Resources); in the theoretical discussion of the article (Supervision); as well as in drafting the article (Writing—original draft); and in the final review of the article (Writing—review & editing).T.G.Y.: in the design and planning of the research within its theoretical framework (Conceptualization); in the theoretical discussion of the article (Supervision); as well as in drafting the article (Writing—original draft); and in the final review of the article (Writing—review & editing).A.S.P.: in the design and planning of the research within its theoretical framework (Conceptualization); in drafting the article (Writing—original draft); and in the final review of the article (Writing—review & editing).M.Á.A.G.: in the design and planning of the research within its theoretical framework (Conceptualization); in the theoretical discussion of the article (Supervision); and in the final review of the article (Writing – review & editing).C.A.V.: in recruitment, data collection, and data cleaning and analysis; collaborating on the literature review (Resources).C.C.R.: in recruitment, data collection, and data cleaning and analysis; collaborating on the literature review (Resources).E.E.: in recruitment, data collection, and data cleaning and analysis; collaborating on the literature review (Resources); and in the final review of the article (Writing—review & editing).

Funding
This research did not receive funding.

Data availability
The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with the ethical principles for research involving human subjects established in the Declaration of Helsinki and the current national ethical guidelines in Chile. In all cases, voluntary informed consent was obtained from the participants before they began the questionnaire, ensuring compliance with ethical principles for research involving human subjects. In addition, the study was approved by the Institutional Bioethics Committee of Andrés Bello University (000047-UNAB). Prior to starting the questionnaire, participants accessed an informed consent letter detailing the study's objectives, participants' rights, and guarantees of confidentiality and anonymity. All participants were adults. Only those who voluntarily agreed to participate were able to proceed with the questionnaire, and they had the option to withdraw at any time without consequences.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 21 November 2025 / Accepted: 31 May 2026
Published online: 11 June 2026

References

1. Albalá–Genol MÁ, Etchezahar E, Gómez–Yepes T, Ungaretti J. Phubbing as a social norm: Effects on fear of missing out (FoMO) and perceived exclusion. Revista Latinoam de Tecnología Educativa. 2025;24(1):65–75. https://doi.org/10.17398/1695-288X.24.1.65.
2. Alfonso-Fuertes I, Alvarez-Mon MA, Sanchez del Hoyo R, Ortega MA, Alvarez-Mon M, Molina-Ruiz RM. Time Spent on Instagram and Body Image, Self-esteem, and Physical Comparison Among Young Adults in Spain: Observational Study. JMIR Formative Res. 2023;7:1–11. https://doi.org/10.2196/42207.
3. American Psychiatric Association. (2013). Diagnostic and Statistical of Mental Disorders - DSM5. In *Guía de consulta de los criterios diagnósticos del DSM-5*. https://www.federaciocatalanatdah.org/wp-content/uploads/2018/12/dsm5-manualdiagnosticoyestadisticodelostrastornosmentales-161006005112.pdf
4. Campbell F, Blank L, Cantrell A, Baxter S, Blackmore C, Dixon J, Goyder E. Factors that influence mental health of university and college students in the UK: a systematic review. BMC Public Health. 2022;22(1):1–22. https://doi.org/10.1186/s12889-022-13943-x.
5. Chao M, Lei J, He R, Jiang Y, Yang H TikTok use and psychosocial factors among adolescents: Comparisons of non-users, moderate users, and addictive users. Psychiatry Res. 2023. https://doi.org/10.1016/j.psychres.2023.115247.
6. Çiçek İ, Şanlı ME, Arslan G, Yildirim M. Problematic Social Media Use, Satisfaction with Life, and Levels of Depressive Symptoms in University Students during the COVID-19 Pandemic: Mediation Role of Social Support. Psihologija. 2024;57(2):177–97. https://doi.org/10.2298/PSI220613009C.
7. Cortez Chávez CV, Palacios-Serna LI, Cubas R, Julca TLR, A. P., Rincón Soto IB. Social Management for The Prevention of Addiction to Social Networks and Anxiety in University Students. Revista de Gestão Social e Ambiental. 2024;18(10):e09084. https://doi.org/10.24857/rgsa.v18n10-129.
8. Coyne P, Csabai BA, Woodruff SJ. A Social Media Give and Take. Int J Social Media Online Communities. 2023;15(1):1–19. https://doi.org/10.4018/ijsmoc.324106.

9. Dhiman A, Varma P. A Differential Study of Compulsive Usage of WhatsApp. Int J Emerg Res Eng Sci Manage. 2023;2(1):25–32. https://doi.org/10.58482/ijeresm.v2i1.5.
10. Dung TTV, Duyen TTN. Social Media Use and Students' Mental Health. VNU J Science: Policy Manage Stud. 2024;40(4):69–82. https://doi.org/10.25073/2588-1116/vnupam.4514.
11. Durao M, Etchezahar E, Albalá Genol MÁ, Muller M Fear of Missing Out, Emotional Intelligence and Attachment in Older Adults in Argentina. J Intell. 2023. https://doi.org/10.3390/jintelligence11020022.
12. Durao M, Etchezahar E, Ungaretti J, Calligaro C. Psychometric properties of the Mobile Phone Dependency Test (TDMB) in Argentina and its relationship with impulsivity. Actualidades en Psicología. 2021;35(130):1–18. https://doi.org/10.15517/ap.v34i129.41963.
13. Etchezahar E, Ungaretti J, Marchiano F. Economic support? They don't really need it. Prejudice towards Latin American immigrants in Argentina. Int J Intercultural Relations. 2022;87:37–41. https://doi.org/10.1016/j.ijintrel.2022.01.007.
14. Etchezahar E, Durao M, Albalá Genol MA, Muller M. Validation of the Perceived Phubbing Scale to the Argentine Context. Behavioral Sciences. 2023;13(2):192. https://doi.org/10.3390/bs13020192.
15. European Commission. (2022). *The Digital Services Act package*. https://digital-strategy.ec.europa.eu
16. Griffiths MD. Disorders due to addictive behaviors: Further issues, debates, and controversies: Commentary to the debate: Behavioral addictions in the ICD-11. J Behav addictions. 2022;11(2):180–5. https://doi.org/10.1556/2006.2022.00025.
17. Gori A, Topino E, Griffiths MD. The associations between attachment, self-esteem, fear of missing out, daily time expenditure, and problematic social media use: A path analysis model. Addict Behav. 2023;141(January):107633. https://doi.org/10.1016/j.addbeh.2023.107633.
18. Huang PC, Latner JD, O'Brien KS, Chang YL, Hung CH, Chen JS, Lee KH, Lin CY. Associations between social media addiction, psychological distress, and food addiction among Taiwanese university students. J Eat Disorders. 2023;11(1):1–9. https://doi.org/10.1186/s40337-023-00769-0.
19. Kanwal A, Munir H, Abrar S. Smartphone addiction, problematic social media use & mental health problems among young adults. Gomal Univ J Res. 2024;40(4):456–66. https://doi.org/10.51380/gujr-40-04-05.
20. Lovibond SH, Lovibond PF. Manual for the Depression Anxiety Stress Scales. Psychol Foundation Australia. 1995;56:42. https://doi.org/10.1037/t01004-000.
21. Lunia S. The Instagram Paradox: Examining Its Role in Shaping Mental Health Among Young Adults–A Comprehensive Analysis. Am J Appl Psychol. 2024;13(3):53–66. https://doi.org/10.11648/j.ajap.20241303.12.
22. Manzoor R, Sajjad M, Shams S, Sarfraz S. TikTok Scrolling Addiction and Academic Procrastination in Young Adults. Pakistan J Humanit Social Sci. 2024;12(04):3290–5. https://doi.org/10.52131/pjhss.2024.v12i4.2592.
23. Miranda S, Trigo I, Rodrigues R, Duarte M Addiction to social networking sites: Motivations, flow, and sense of belonging at the root of addiction. Technol Forecast Soc Chang. 2023. https://doi.org/10.1016/j.techfore.2022.122280.
24. Mohamed EF, Mohamed AE, Youssef AM, Sehlo MG, Soliman ESA, Ibrahim AS Facebook addiction among medical students: prevalence and consequences. Middle East Curr Psychiatry. 2025. https://doi.org/10.1186/s43045-025-00501-5.
25. Montag C, Lachmann B, Herrlich M, Zweig K. Addictive Features of Social Media/Messenger Platforms and Freemium Games against the Background of Psychological and Economic Theories. Int J Environ Res Public Health. 2019;16(14):2612. https://doi.org/10.3390/ijerph16142612.
26. Oldemburgo de Mello V, Cheung F, Inzlicht M Twitter (X) use predicts substantial changes in well-being, polarization, sense of belonging, and outrage. Commun Psychol. 2024. . https://doi.org/10.1038/s44271-024-00062-z.
27. Primi C, Fioravanti G, Casale S, Donati MA. Measuring problematic facebook use among adolescents and young adults with the bergen facebook addiction scale: A psychometric analysis by applying item response theory. Int J Environ Res Public Health. 2021;18(6):1–15. https://doi.org/10.3390/ijerph18062979.
28. Sahebi S, Formosa P. Social media and its negative impacts on autonomy. Philos Technol. 2022;35(3):70. https://doi.org/10.1007/s13347-022-00567-7.
29. Stănculescu E, Griffiths MD Social media addiction profiles and their antecedents using latent profile analysis: The contribution of social anxiety, gender, and age. Telematics Inf. 2022. . https://doi.org/10.1016/j.tele.2022.101879.
30. Tutgun-Ünal A, Deniz L Development of the Social Media Addiction Scale. Online Acad J Inf Technol. 2015. . https://doi.org/10.5824/1309-1581.2015.4.004.x.
31. Villegas Dominguez J, Muñoz Bautista B, Gonzalez David E, Vásquez Gárate R, Alcantara U, Pavan Gallardo S, Reyes Hernández J, Garduza CC, A., Celedonio Márquez, F, M. Self-esteem associated with the use of social networks in university students. Eur J Pub Health. 2022. https://doi.org/10.1093/eurpub/ckac131.255.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.